



Where correction helps — and what the mean hides

Coupled Active Inference on a Frozen Transformer Prior: A Risk-Aware Residual Agent for Non-Equilibrium Regimes

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A frozen language model. A small adaptive memory. Controlled tests of correction and unintended interference.



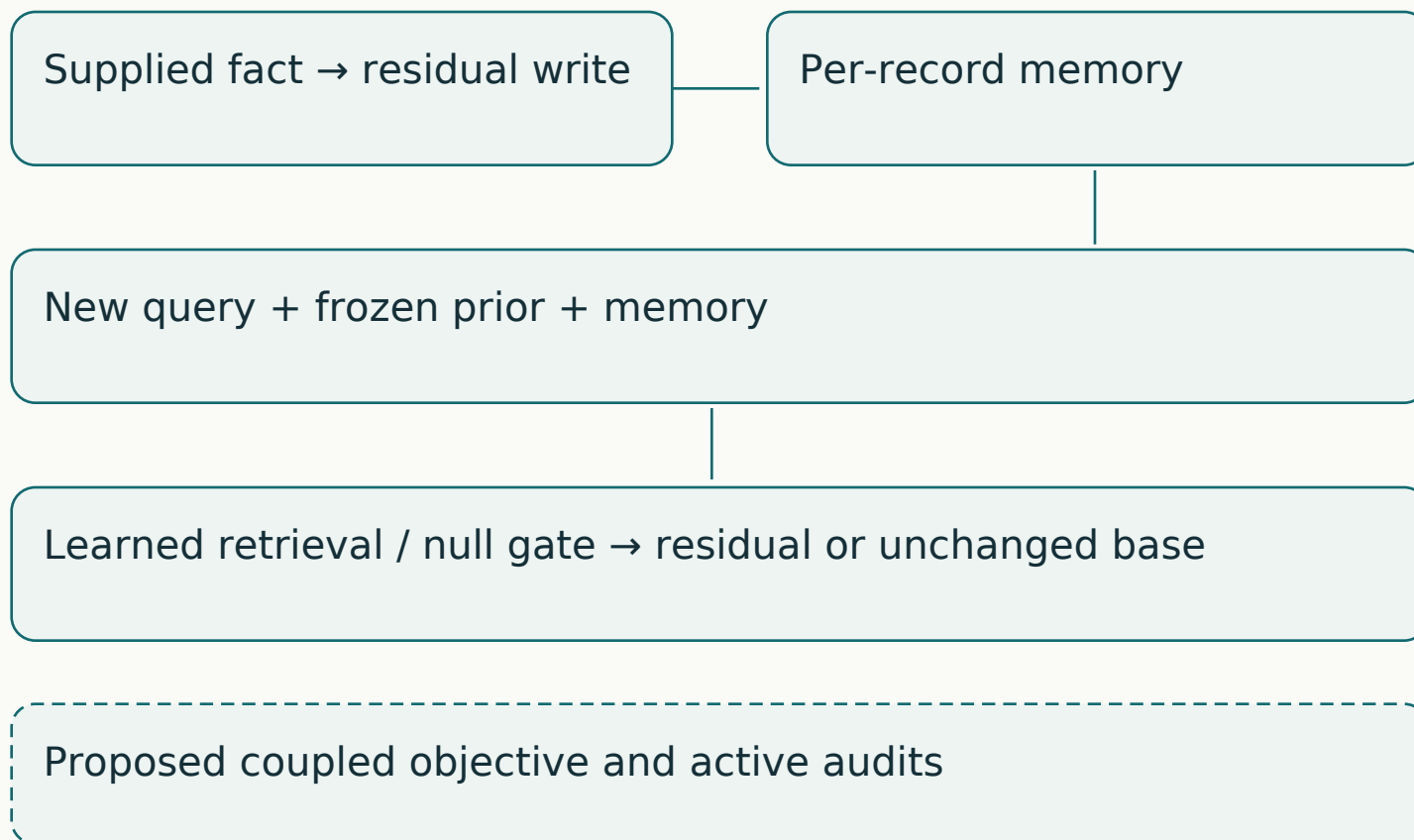
The system we actually built

Frozen GPT-2 base (~124M parameters)

3,348,228 trained reader/controller parameters (~2.7%)

Separate per-record memory state

Gate acceptance $\neq$ nonzero residual $\neq$ changed answer



Interfaces are architectural analogies. Freezing weights proves neither a Markov blanket nor universal preservation.



Retention and rejection compete

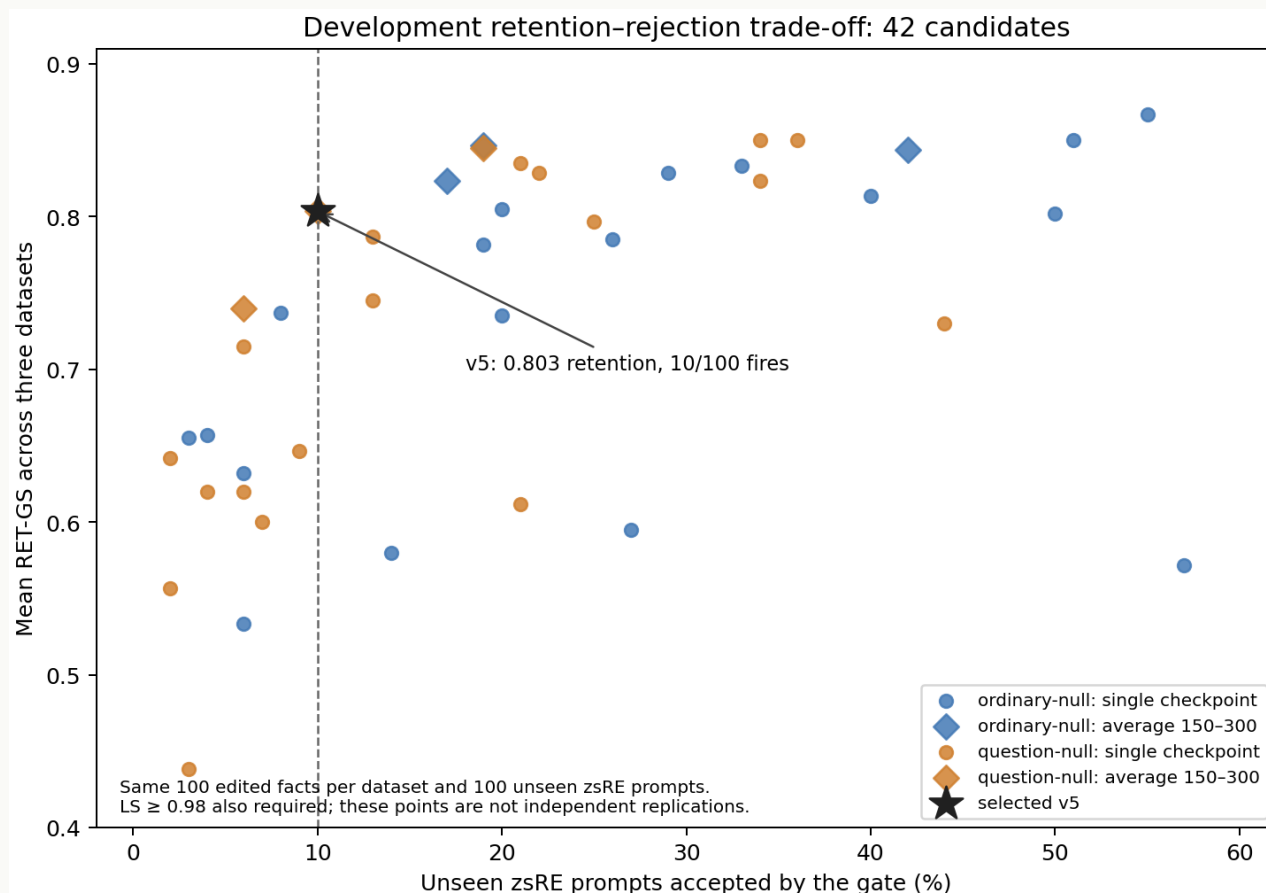
42 candidates · 15 admissible

Selected v5 macro RET-GS:
0.8033

zsRE .98 · CounterFact .82 ·
MQuAKE .61

zsRE unseen gate
acceptance: 10/100

Fixed-candidate Wilson 95%:
5.52–17.44%



Development selection, not confirmation. The interval is not selection-adjusted and does not certify a rate below 10%.



A small average can hide a large local loss

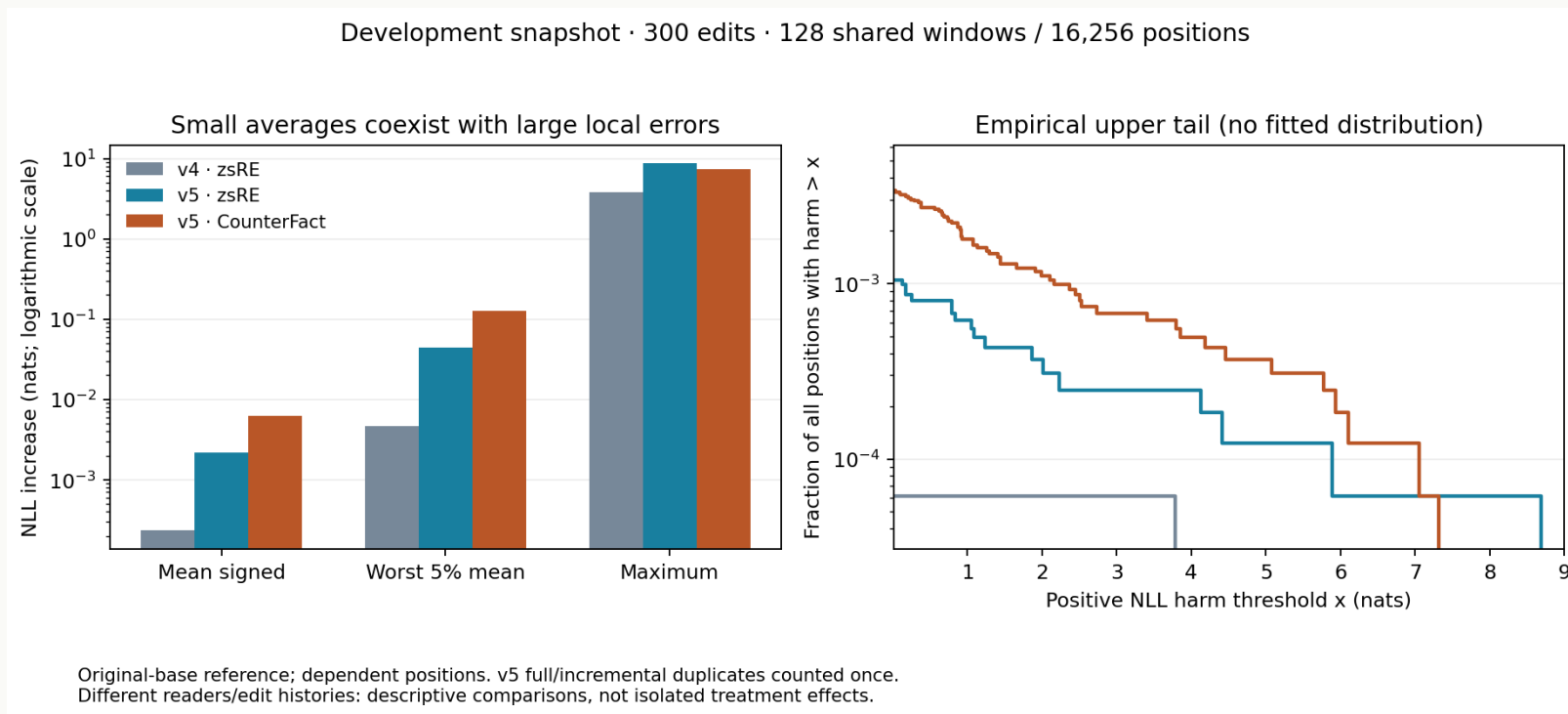
zsRE mean signed Δ NLL:
.00220 nats

ES95 positive harm: .04396
nats

Worst position: 8.69 nats

17 of 16,256 positions exceed
.1 nat

+8 nats $\approx$ 3,000 $\times$ less likely
target token



Rare concentrated harm; no power-law or tail-exponent claim. Positions are dependent; saved profiles are not independent replications.



Complete validation sharpens the picture

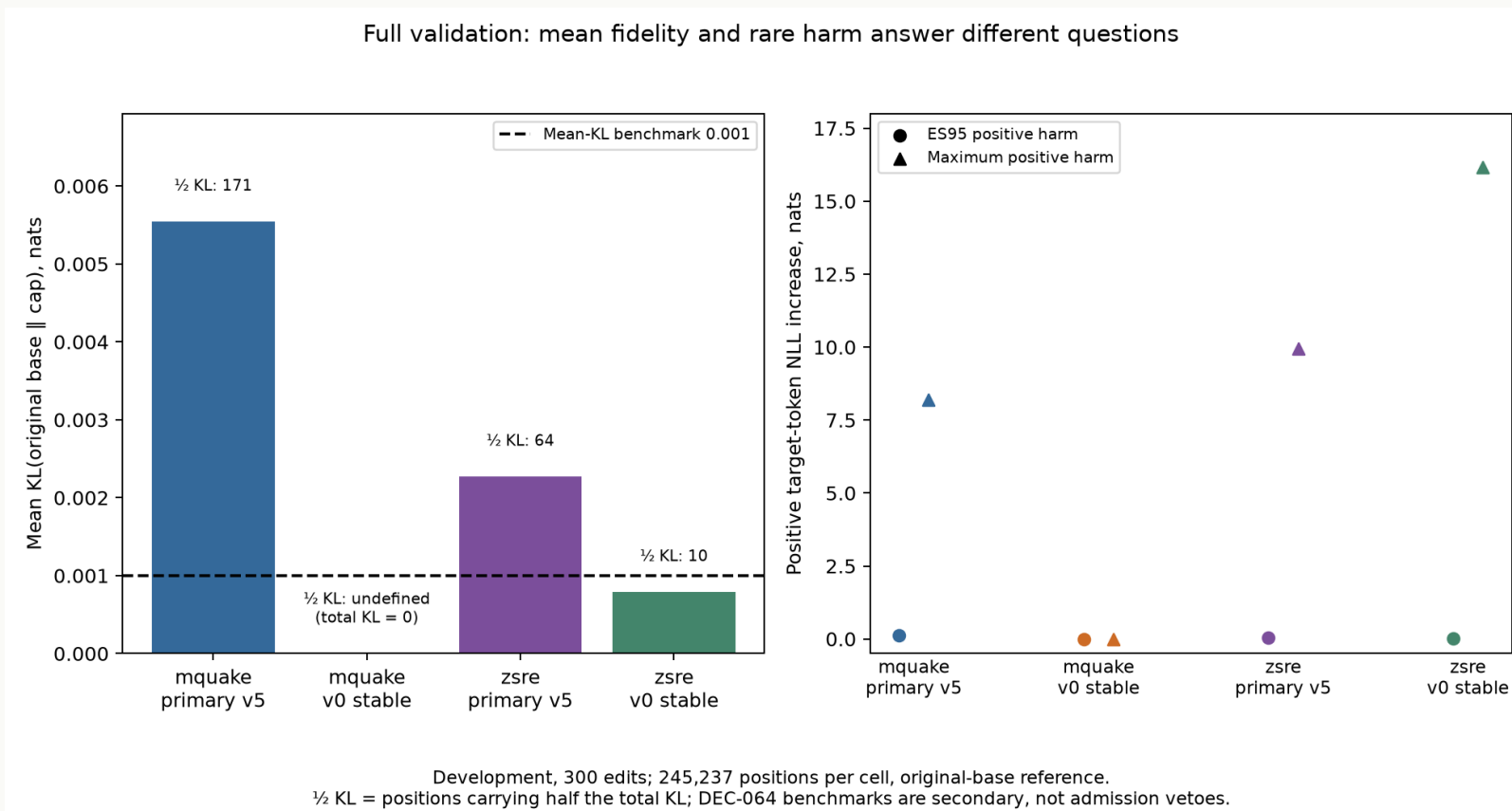
Learned MQuAKE: mean KL
.005544

Learned zsRE: mean KL
.002270

Both exceed the .001 KL
benchmark

zsRE v0: mean KL .000789

Yet its worst Δ NLL is 16.159
nats



DEC-064: cap benchmarks are secondary labels, without an admission veto. Continued-base certification is separate. Fixed windows are not iid.



Three failures changed the investigation

Reader fires too widely → explicit null / rejection training

Occupancy and source confounding → common-population qualification

Benign-looking averages → per-position tails and complete validation

No randomized isolation of each repair. Investigators chose these audits; the agent did not perform expected-free-energy action selection.



A small, specified κ intervention

Answer surprisal: $(1 - p^\kappa) / \kappa$

$\kappa \rightarrow 0$ gives ordinary $-\log p$

$\kappa = .2$ and $.5$; ordinary and clip-2 controls

Fixed checkpoint average: 150–300

Only the answer term is bounded; not the whole objective or all gradients



The κ pilot did not meet its success rule

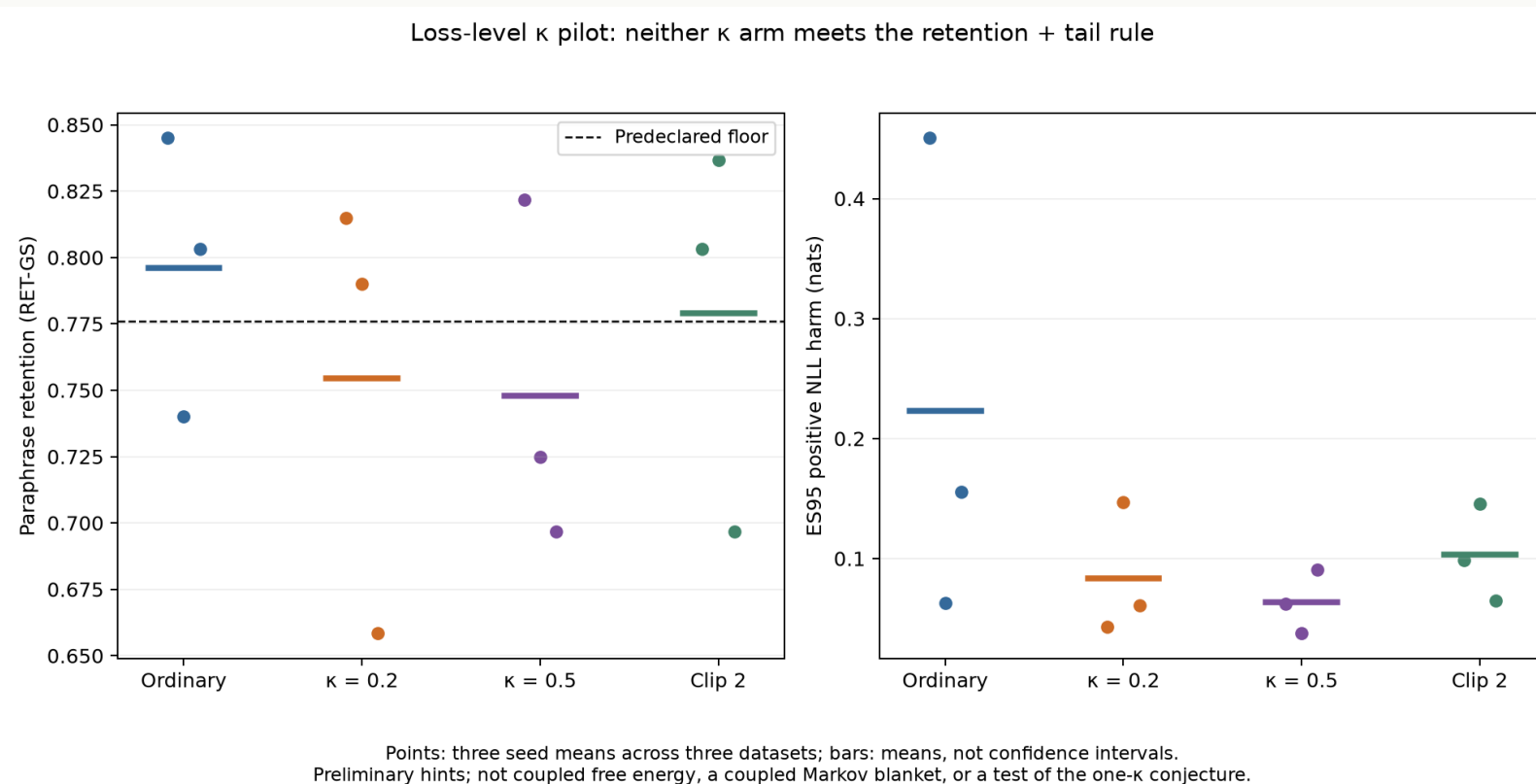
RET-GS: ordinary .7961

κ .2: .7544 · κ .5: .7478

Predeclared retention floor:
.7761

Both κ arms fail retention and
tail separation

Clip2 meets the retention
floor; fails separation and
adds a CounterFact false fire



Preliminary hints at what the architecture could provide; not the coupled free energy (no coupled expectation or changed inference distribution), not a coupled Markov blanket, and not a test of the one- κ conjecture that porosity and interference are the same parameter.



Correct answers can conceal probability harm

Each tested dataset retains
20/20 old exact answers at
observed checkpoints

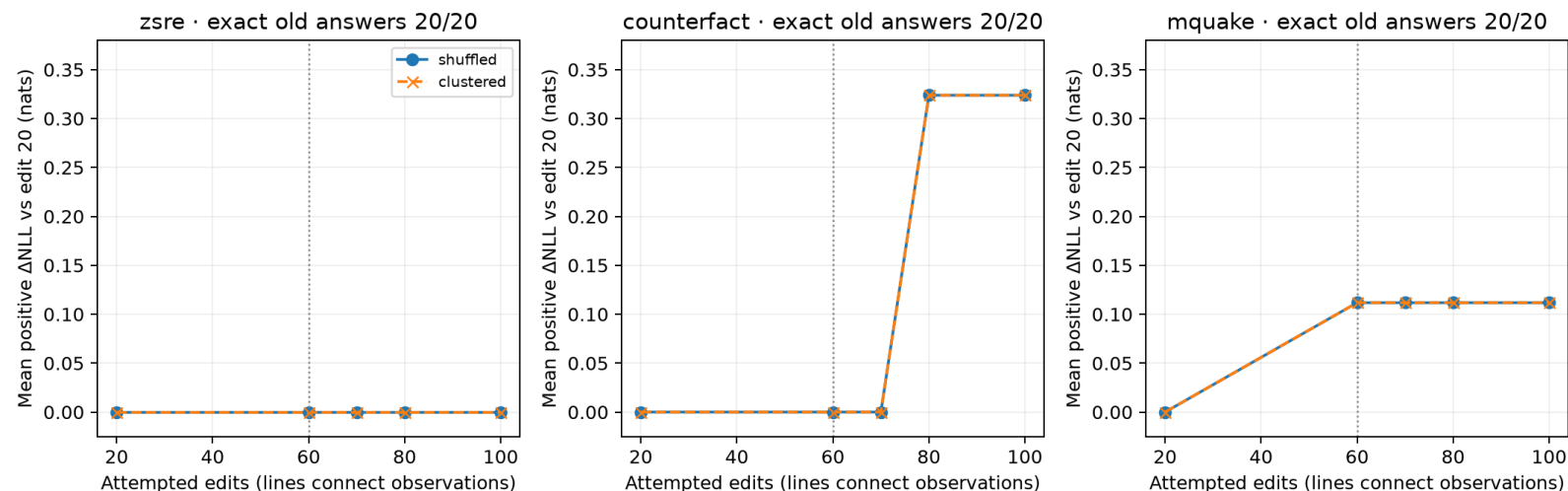
CounterFact at edit 100:

Mean positive ΔNLL .32379

Maximum 12.53188 nats

Harm persists through the
last observation

Old-fact probability harm persisted through the observation window



Development: one seed, two tested schedules; overlapping curves are observed equality.
CounterFact: 20 updates observed after first detected harm; MQuAKE: 40. Recovery beyond edit 100 is unknown.

One seed, two schedules, measured probes only. Recovery is unobserved, not impossible or permanent; equal schedules do not prove order invariance.



What independent confirmation can establish

zsRE: 120 core cells

CounterFact: 120 core cells

MQuAKE: 45 core cells at 300 edits

285 core + 45 optional cells

MQuAKE core: primary v5, random reader, v0 stable

75 MQuAKE cells are prospectively omitted, not measured zeros. Keep 63 intervals, including 21 unavailable MQ1000 intervals. S1 is not exact-v5 compute matched.



Headroom is conditional; October 9 is the stop

431.31 expected process-hours

646.96 at every cell ceiling

750 shared process-hour cap

≈227.30 elapsed hours at assumed 1.65× throughput

October 10–14: locked-data analysis, figures, rehearsal



What would distinguish a coupled objective?

Measured: concentrated harm and loss-level trade-offs

Proposed: coupled expectations and correlated-regime adaptation

Test: improve the correction-tail-harm trade-off beyond clipping and better rejection training, while preserving retention

Preliminary hints, not coupled free energy or a test of the one- κ conjecture. The present null neither refutes nor confirms the broader programme.